

```
%Name      : Busisiwe Rose Kubeka
%Module     : SME3701_Assessment_1
%Date       : 30 April 2026
%GitHub Link : https://github.com/55394949/55394949/edit/main?README.md
```

```
clc; clear; close all;
```

```
%Given
```

```
m = 5;
k = 2000;
x0 = 0.02;
v0 = 0;
```

```
%Q1.1 Natural Angular Frequency
```

```
wn = sqrt(k/m);
fprintf('Natural Angular Frequency=%.2frad/s\n',wn);
```

Natural Angular Frequency=20.00rad/s

```
%Q1.2 Displacement equation
```

```
wn = sqrt (k/m);
t = 0:0.01:5;
%Displacement equation
x = x0 * cos(wn * t);
disp ('Displacement values x (t):');
```

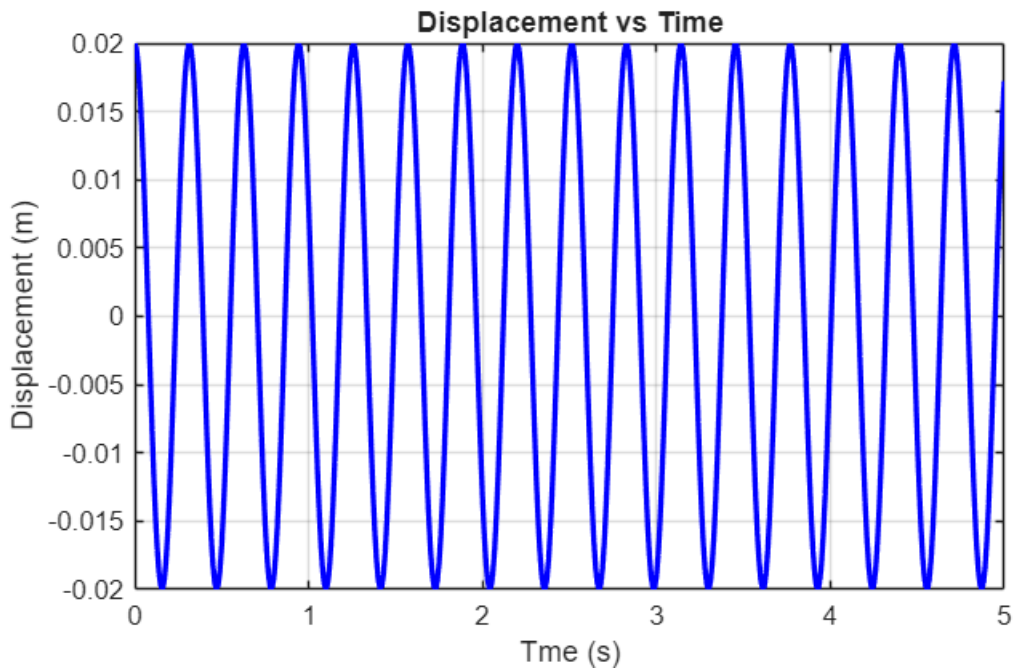
Displacement values x (t):

```
disp (x);
```

0.0200 0.0196 0.0184 0.0165 0.0139 0.0108 0.0072 0.0034 -0.0006 -0.0045 -0.0083 -0.0120

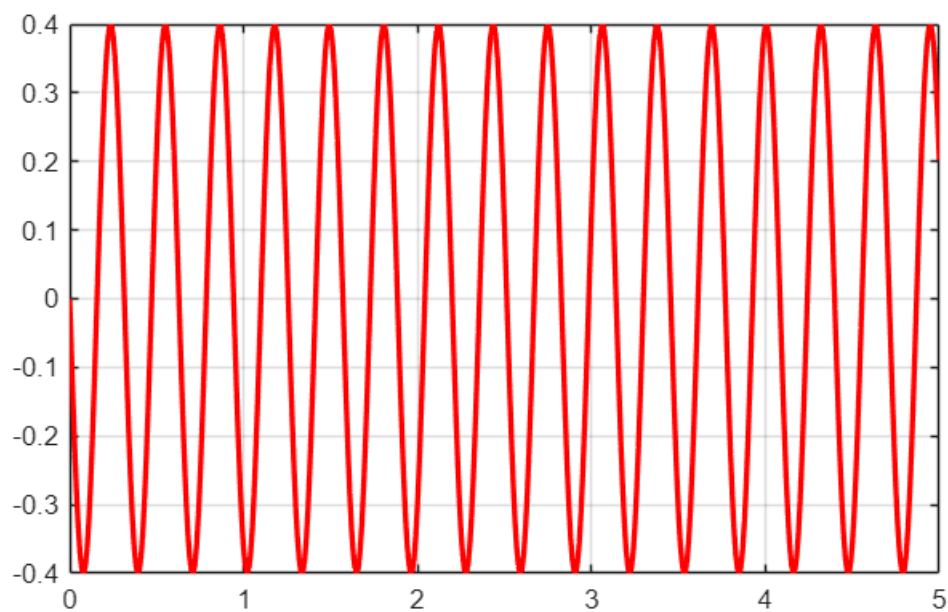
```
%Q1.3 Displacement plot
```

```
wn = sqrt (k/m);
t = 0:0.01:5;
x = x0 * cos (wn * t);
plot (t,x,'b', 'linewidth',2);
grid on;
xlabel('Time (s)');
ylabel('Displacement (m)');
title('Displacement vs Time');
```



```
%Q1.4 Velocity plot
%Given
m = 5;
k = 2000;
x0 = 0.02;
v0 = 0;

wn = sqrt(k/m);
t = 0:0.01:5;
% velocity equation
v = -x0 * wn * sin(wn * t);
plot (t,v,'r', 'linewidth',2);
grid on;
```



%Q1.5 Comments

```
disp ('nature of motion: simple harmonic');
```

nature of motion: simple harmonic

```
disp ('zero damping meaning: No energy loss');
```

zero damping meaning: No energy loss

```
disp ('velocity is 90 degrees out of phase with displacement');
```

velocity is 90 degrees out of phase with displacement

```
disp ('Velocity is maximum at zero displacement');
```

Velocity is maximum at zero displacement